# Per-Group Error, Not Total MSE: Fine-Tuning Vision-Language-Action Models for 11-DoF Mobile Manipulation

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## Problem Statement

Vision-Language-Action models (VLAs) take a camera image and a language instruction and output a sequence of robot actions. They are trained on large robot teleoperation datasets and evaluated offline by computing the **prediction error (MSE)** on **held-out data**.

- Most VLAs are pretrained on **fixed arms with 6-7 DoF**
- The **Toyota HSR has 11 DoF**: arm (5), gripper (1), head (2), wheeled base (3).
- These groups have **different scales and dynamics**: gripper is binary, base is continuous.
- **Aggregate MSE** collapses these 11 dimensions into a single number.

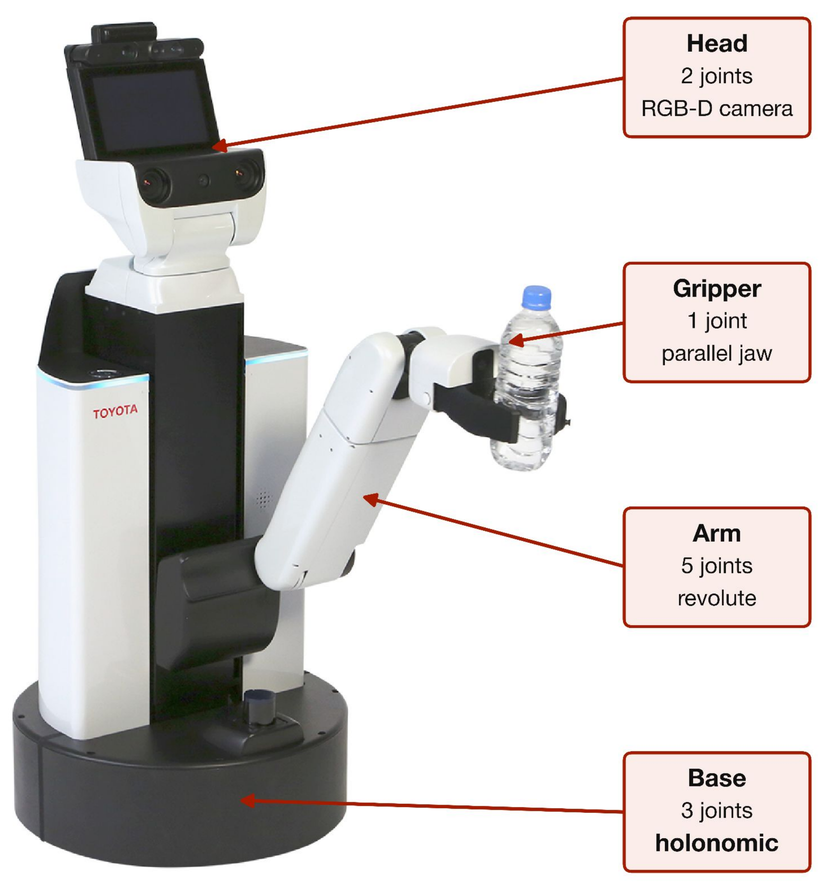
⇨ **Does the lowest-MSE checkpoint really pick the best policy for the robot?**

## Contribution

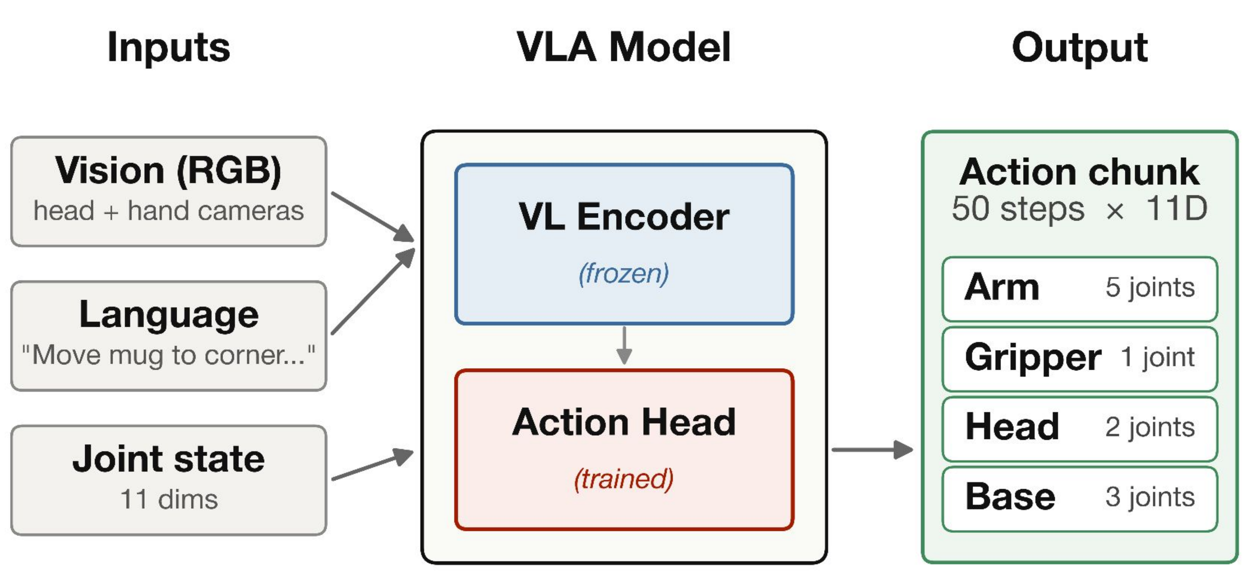
⇨ **Lowest total MSE does not pick the best policy. Per-group MSE is a better signal than total MSE.**

- **Per-group MSE decomposition** by joint group (arm / gripper / head / base) as a checkpoint-selection signal.
- Two-phase fine-tuning of **SmolVLA (450M)** [1] with native 11-DoF output.
- Comparison against **$\pi_{0.5}$ (3.3B)** [2] baseline + expert-only fine-tuning.
- **60 real-robot trials** validate the offline finding.
- **1st place / 36 teams** in the AIRoA VLA Pipeline Competition (ICRA 2026) [3]

## Method Overview




*Toyota Human Support Robot (HSR). 11-DoF spread across four functionally distinct joint groups.*




*Vision-Language-Action pipeline. The vision-language encoder stays frozen during fine-tuning, only the action head is trained.*

Aggregating MSE over the 11-DoF action vector collapses groups with different scales and dynamics. We decompose into four functional groups G = {arm, gripper, head, base} and compute:

$$\mathrm{MSE}_g = \frac{1}{|g|} \sum_{j \in g} (\hat{a}_j - a_j)^2, \quad g \in \mathcal{G}$$

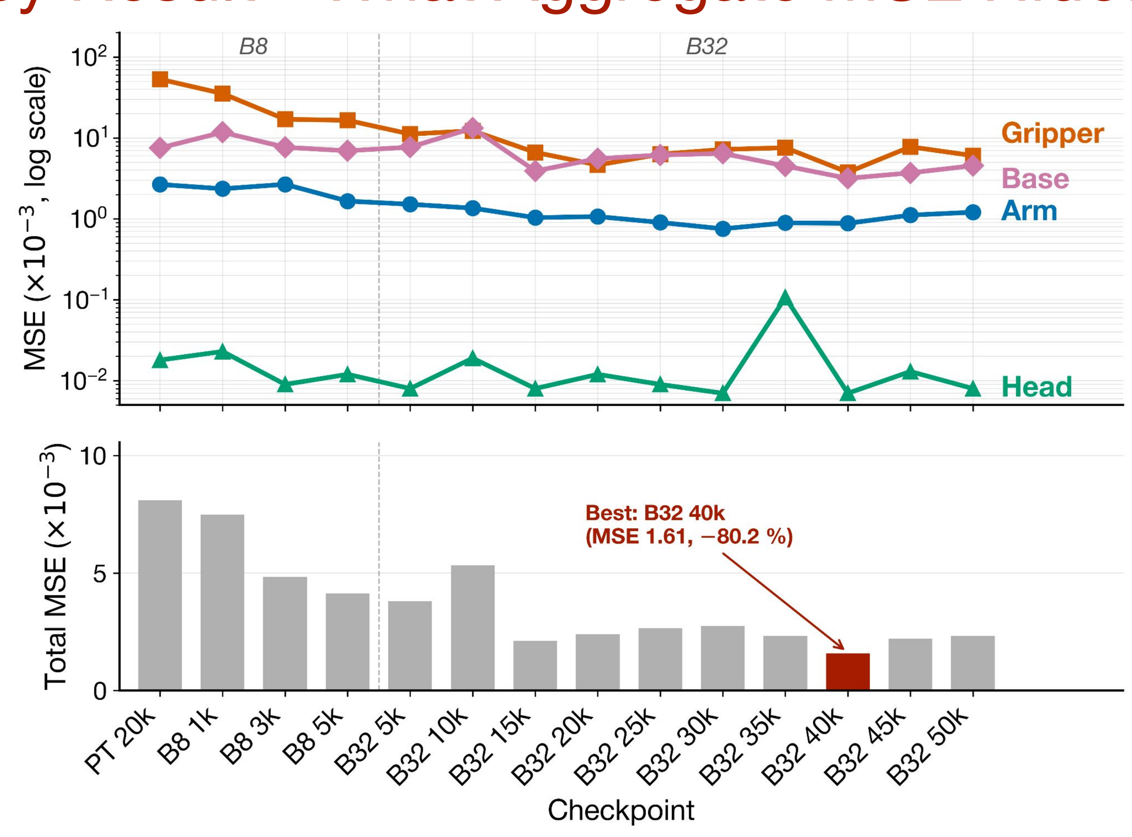
- Reveals which groups benefit from fine-tuning and which remain bottlenecks.
- Hidden by aggregate metrics.
- Two uses: (a) **checkpoint selection** for deployment, (b) **diagnostic** of which joint group is under-trained.

## Key Result - What Aggregate MSE Hides

### SmolVLA (450M)[1]

HSR-SmolVLA (40k):
Total MSE = 1.61
−80% vs. pretrained checkpoint

Arm 0.88 · Gripper 3.77 · Head 0.01 · Base 3.18 ($\times10^{-3}$)
Gripper converges 10× faster than the base.




*Per-joint-group MSE over SmolVLA training. The gripper falls fast, the base converges last and sets the ceiling on total MSE.*

### $\pi_{0.5}$ (3.3B)[2]

Expert-only beats baseline on total MSE (0.95 < 1.04)
but arm-MSE doubles (0.30 → 0.59)

Baseline 80k: 0.30 / 4.31 / 0.05 / 1.85
Expert-only 3k: 0.59 / 2.21 / 0.06 / 1.72 ($\times10^{-3}$)

**Aggregate MSE collapses the four groups. Per-group MSE shows which one matters for deployment.**

## From Offline to Real World

### Real-Robot Validation (n=60)

| Model | Total | Arm | Robot |
|---|---|---|---|
| $\pi_{0.5}$ 80k | 1.04 | **0.30** | **4.00 / 4** |
| $\pi_{0.5}$ exp. 3k | **0.95** | 0.59 | 3.75 / 4 |
| SmolVLA 40k | 1.61 | 0.88 | 3.50 / 4 |

MSE values × $10^{-3}$. Robot score: 4-point rubric, mean over 20 trials per model.

Arm-MSE matches the robot ranking.
Total MSE does not.

Mann–Whitney U, $p \leq 0.010$

### Validation at Scale

**1st place out of 36 teams**

- 60% mean complete-task success rate
- 5 of 6 tasks reach ≥ 60% complete-task success rate

| Tasks (6) | Success rate |
|---|---|
| Public (3) | 70% / 80% / 80% |
| Private (3) | 70% / 60% / 0% |

1 private task: 0%, object not in the training distribution.

### Take-Home

- Report per-group MSE alongside total.
- The dominant group depends on model and regime.
- Keep fine-tuning short when task data is scarce.

**Full Paper** **Code**

[1] Shukor, Mustafa, et al. "Smolvla: A vision-language-action model for affordable and efficient robotics." arXiv preprint arXiv:2506.01844 (2025).
[2] Black, Kevin, et al. "$\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization." 9th Annual Conference on Robot Learning (2025).
[3] AIRoA. "AIRoA 10k Dataset: A Large-Scale Mobile Manipulation Dataset for VLA Pipelines." ICRA 2026 Workshop: From Data to Decisions (2026).